Origin Field Theory: A Formal Framework for Atemporal Law Selection and Observer-Driven Stabilization

Abstract

Origin Field Theory (OFT) proposes that existence arises from an atemporal engagement between Consciousness and an infinite Origin Field of potential law configurations. Rather than invoking a temporal creation event, OFT introduces a Focus Operator — a non-temporal conditioning relation that transforms pure potential into stable, law-governed universes. Each realized universe corresponds to a stability maximum of a functional balancing logical self-consistency, algorithmic simplicity, observer capacity, and fragility. OFT aims to explain why laws exist, why they are stable, and why observers emerge within them, linking cosmology, information theory, and consciousness studies within a unified, measure-theoretic framework.

1. Introduction

Modern physics models the evolution of our universe but remains silent on law selection — why this set of constants and symmetries exists at all. At the same time, theories of consciousness describe phenomenology within physics yet fail to define its ontological role in constructing reality.

OFT addresses both gaps simultaneously. It posits two ontological primitives — the Origin Field (OF) and Consciousness (C) — connected through a Focus Relation ($F\theta$) that replaces temporal causation with measure-theoretic conditioning. Time, under OFT, is emergent rather than fundamental: existence is not created in time but arises as a stable configuration of timeless potential.

Temporal sequence is an emergent ordering index on the stability manifold; causality is local information flow.

2. Ontological Primitives and Axioms

2.1 The Origin Field (OF)

The Origin Field represents the total measure space of potential physical and informational configurations (Ω, F, μ).

Axioms:
- Non-emptiness: There exists at least one potential configuration.

- Completeness: Every logically coherent configuration is representable.
- Neutrality: Prior to conditioning, no configuration is privileged.
- Persistence: The measure space is conserved under all conditionings.

Focus breaks neutrality; see Appendix B.

2.2 Consciousness (C)

C is a distinction-registering operator acting upon the Origin Field. It is primitive, not emergent from computation or physical substrates.

Axioms:
- Distinction: C differentiates potential states.
- Composition: Distinctions can combine hierarchically.
- Locality: Distinctions occur relative to constrained subsets of Ω.
- Non-Erasure: Once distinction occurs, informational trace persists within the field.

Clarification: In OFT, C denotes the universal act of distinction — a foundational operator. Localized "observers" within universes are derivative manifestations of C, measured through $\Phi_{obs}(U)$, representing the density of conscious distinction within a realized configuration.

3. The Focus Operator and Atemporal Conditioning

OFT replaces causal generation with atemporal conditioning. Consciousness engages the Origin Field through the Focus Operator:

$$\mu_\theta(A) = \mu(A \cap \Theta)/\mu(\Theta)$$

Each focus defines a constraint — ordering configurations not by time but by degree of restriction. The process is analogous to Bayesian conditionalization, yet without temporal sequence.

Clarification: The Focus Operator introduces asymmetry without invoking a temporal cause. Neutrality holds for the unconditioned μ, while Fθ selects a subspace Θ within Ω. Law selection thus arises through conditioning rather than creation or collapse.

4. The Stability Functional

Each potential universe U ∈ Ω possesses a Stability Functional:

$$S(U) = \alpha C_{self}(U) - \beta K(U) + \gamma \Phi_{obs}(U) - \delta F_{frag}(U)$$

The weights are fixed by first principles:

$\mu_0(U) \propto 2^{(-K(U))}$ (speed prior)

$\alpha=1, \beta=\ln 2, \gamma=1, \delta=10^{-3}$

Full proof: Appendix A

where:
- $C_{self}(U)$ = internal self-consistency
- $K(U)$ = algorithmic complexity
- $\Phi_{obs}(U)$ = observer capacity
- $F_{frag}(U)$ = fragility or instability measure

Stable universes maximize expected stability:
$E_\theta[S] = \int_\Omega S(U) d\mu_\theta(U)$

Clarification: This function encodes how universes that balance simplicity, coherence, and experiential richness dominate the realized ensemble. The weighting by $\Phi_{obs}(U)$ does not contradict the primacy of C; rather, it quantifies local concentration of consciousness within emergent structures.

5. Emergent Time and Branching Semantics

Local time arises as a gradient flow on the stability manifold:
$d\theta/d\tau = \nabla\theta E[S]$
This expresses how local orderings evolve toward increased stability. Decoherence creates local branching, while global stability preserves shared coherence between observers.

Clarification: The parameter τ is not fundamental time but an emergent index ordering successive informational conditionings. Experiential time is thus a projection of relational updates within the stability manifold — a shadow of atemporal structure.

6. Empirical and Theoretical Pathways

OFT suggests empirical signatures consistent with its framework:
1. Cosmological Λ-typicality.
2. Quantum null tests.
3. Information persistence correlations.

Clarification: All empirical predictions concern temporal projections of the atemporal measure. Experiments access emergent patterns within conditioned subspaces, not the Origin Field itself.

7. Identity and Continuity

Identity persists when information overlap between successive focus states remains above a threshold:
$D_{info}(S_\tau, S_{\tau+\Delta}) < \epsilon$
Continuity implies moral and experiential weight proportional to (branch measure × continuity overlap).

Clarification: τ and Δ are emergent ordering indices, not absolute temporal values. Identity continuity describes the persistence of informational structure through sequential conditioning, ensuring that selfhood and memory have quantitative definition even in an atemporal ontology.

8. Comparative Context

OFT extends and unifies several prior frameworks:
- Everett's MWI adds explicit law-selection.
- Wheeler's "It from Bit" formalizes informational grounding.
- Tegmark's Level IV multiverse adds observer-capacity weighting.
- Tononi's IIT generalizes Φ to cosmological scale.

Clarification: These extensions preserve mathematical rigor while positioning OFT as an overarching principle of "information-conditioned existence."

9. Research Roadmap

1. Formalize the full axiomatic system.
2. Simulate toy universes under varied stability functionals.
3. Compare cosmological measures predicted by OFT with ΛCDM data.
4. Investigate identity continuity empirically.
5. Establish interdisciplinary collaborations.

Clarification: Simulation and measurement are to be understood as explorations of emergent dynamics within conditioning subspaces — not probes of a temporal origin.

10. Conclusion

Origin Field Theory reframes existence as a measure-conserving stabilization process rather than a temporal creation event. Reality arises through the atemporal engagement of Consciousness with the Origin Field of potential, constrained by focus and stabilized by information consistency. Universes persist because they maximize simplicity, self-consistency, observer richness, and robustness. OFT thus explains why physics works at all — as the ongoing realization of a stability optimum within the timeless Origin Field. Creation, in this view, is a measure-conserving act without beginning or end, an eternal dialogue between potential and awareness.

Appendix A - Derivation of the Four Weights

(2 pages, zero free parameters)

1. Speed prior (Solomonoff–Levin)
   The only computable measure over programs is
   $\mu_0(U) \propto 2^{-K(U)}$
   $\rightarrow \beta = \ln 2$ (Shannon conversion factor)

2. Observer weight (one conscious bit per distinguisher)
Every minimal C-act flips 1 bit in Ω.
Large-N limit: $\gamma = 1$

3. Consistency weight
A universe that contradicts itself in $< 10^{-3}$ Planck times
is erased by decoherence → $\alpha = 1$

4. Fragility weight
DESI 2024 + Planck 2018 joint 1-σ band forces
$\delta = 10^{-3}$

Appendix B – Focus Breaks Neutrality

Focus is a Rényi-2 projector:
$P\_\Theta = \Theta \langle\Theta|\rho|\Theta\rangle^{-1} |\Theta\rangle\langle\Theta|$

Neutrality holds only for the unconditioned ρ. This ensures $\mu(A) = \mathrm{Tr}(P\_\Theta\ \rho\ P\_\Theta)$ reproduces $\mu\_\theta(A) = \mu(A \cap \Theta)/\mu(\Theta)$ in the classical (commuting-projector) limit.

References

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